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EXPERIMENT 17

PROLOG PROGRAM – SUM OF INTEGERS FROM 1 TO N

Aim

To write and execute a **Prolog program to find the sum of integers from 1 to N** using recursion.

Algorithm

1. Start the program.
2. Read the value of N.
3. If $N = 0$, return the sum as 0.
4. Otherwise, calculate the sum of integers from 1 to $N-1$.
5. Add N to the obtained sum.
6. Continue the recursive process until N becomes 0.
7. Display the final sum.
8. Stop the program.

Prolog Program

% Sum of integers from 1 to N

sum_to_n(0, 0).

sum_to_n(N, Sum) :-

N > 0,

N1 is N - 1,

sum_to_n(N1, Sum1),

Sum is N + Sum1.

Input

?- sum_to_n(10, Sum).

Output:

Sum = 55 ? |

Result

Thus, the **Prolog** program to calculate the sum of integers from 1 to N was successfully **written and executed**, and the correct sum was obtained.